FIN355: Equity Securities

<u>Group-Based Assignment</u>

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Table of Contents

Question 1 ..... **2**

Part (a) - Singapore as an Investment Destination 2

Part (b) - Appraisal of the Energy and Industrials Sector in Singapore 5

Part (c) - Competition Analysis 8

Ratio Analysis 11

Financial Ratios 11

Valuation Ratios: 12

Company of Choice for Further Analysis 13

Question 2 ..... **15**

Part (a) - Singapore as an Investment Destination 15

Part (b) – Projected Growth Rates of Company (High & Normal) 16

Part (c) – Discounted Cash Flow Valuation Method 19

References **30**

Question 1

Part (a) - Singapore as an Investment Destination

Despite the heightened uncertainty in the global economy following Trump's inauguration, Singapore remains an attractive investment destination. With reference to Figure 1, Singapore's economy grew by 4.4% in 2024, up from 1.8% in 2023, largely driven by manufacturing (electronics clusters) and finance and insurance, with Q4 GDP rising 5.0% year-on-year (MTI, 2024) (Figure 1.1).

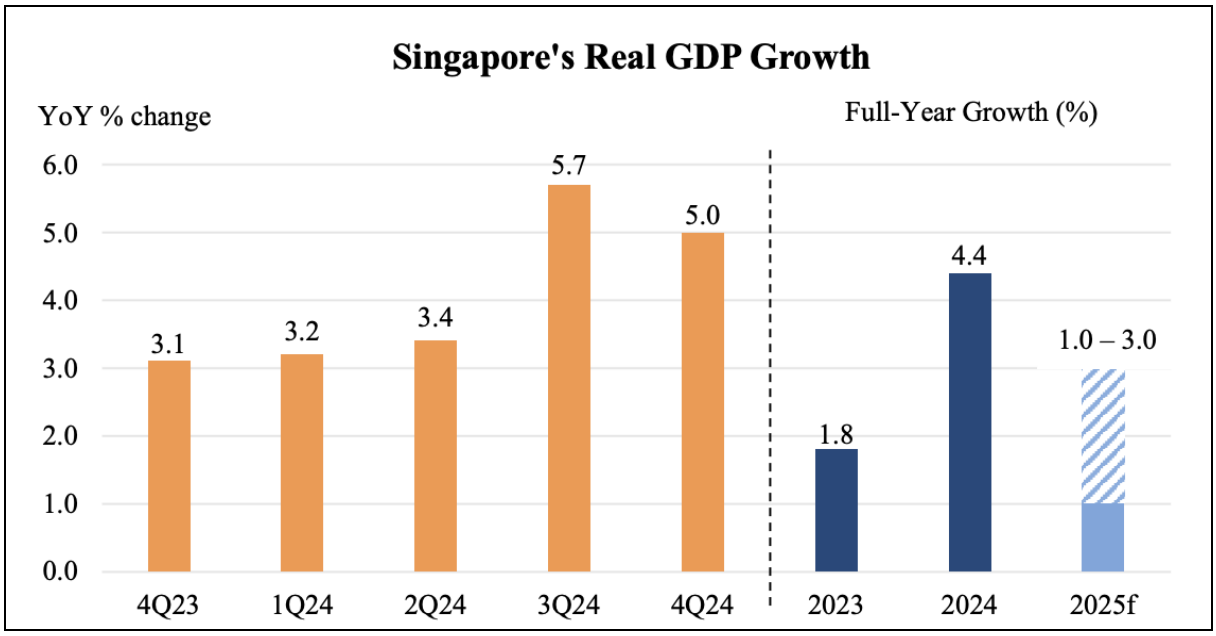

Figure 1.1 - Singapore's Past Economic Performance and Forecast

Due to the upturn in the global electronics cycle, the electronics cluster of the manufacturing sector experienced a notable expansion. This is significant because semiconductors contribute approximately 7% of Singapore's GDP and 40% of its manufacturing output (Cheng & Okawa, 2025). This is attributed to Singapore's advantageous infrastructure, central location in Southeast Asia, and efficient shipping routes (Medina, 2024), which attract international investments and promote efficient supply chains. Additionally, the shifts in global and domestic financial market sentiments led to elevated trading activity, which in turn grew the finance & insurance sector. On the other hand, a shift in consumer behaviours (i.e. rise in overseas travel expenditures) negatively impacted the retail trade and food & beverage services. In 2025, GDP growth is predicted to moderate to 1.0% and 3.0% after considering both the domestic and external environments.

Global tensions between the United States and China pose risks to Singapore's economy (Kok et al., 2025), given that they are Singapore's major trading partners (MTI, 2024). In 2024, Mainland China was Singapore's largest trading partner, with $94.4 billion in exports and $75.7 billion in imports. This was followed by the United States, with $68.7 billion in imports and $69.9 billion in exports. Trade frictions between the two major players, along with the US's protectionist policies, may raise production costs and reduce worldwide investment and trade, increasing global economic uncertainty. While U.S. GDP growth may slow due to a deceleration in private consumption and China faces challenges with weaker merchandise exports and investment growth (MTI, 2024), Singapore's position as a strategic trade hub leader, sovereign credit rating of AAA, and strong fiscal policies, helps to mitigate the downside risks of investments in Singapore.

Despite moderated estimated GDP growth and heightened uncertainty in the global economy in 2025, Singapore's structural strengths and government support ensure stability and upside potential for equity investments over the next 12 months.

Singapore's recognition as a major Asian trading and financial hub has been supported by its political stability, low levels of corruption, and open public institutions (EDB, n.d.). According to the World Bank Institute's annual Worldwide Governance 2023 indicators (see Table 1), Singapore's quality of governance is ranked favourably due to minimal corruption and business-friendly legislative and regulatory frameworks. Similarly, Singapore is ranked first out of close to 200 countries in the Heritage Foundation's annual Index of Economic Freedom Survey (2025), attributing to the country's freedom in trade, business, investment, monetary and financial freedom. This creates a stable investment environment with minimal political and economic risk, thereby attracting investors. Singapore's status as a politically-sound country, along with its predicted moderation in GDP growth, may also attract risk-averse investors who intend to preserve capital during a time of uncertainty in the global economy.

Table 1 - Singapore's Performance in the Worldwide Governance Indicator

Indicator	**Percentile Rank (1 - 100)**
Political Stability and Absence of Violence/Terrorism	97.156
Government Effectiveness	100
Regulatory Quality	100
Rule of Law	98.113
Control of Corruption	98.113

Singapore has a AAA credit rating from major agencies such as Fitch, S&P and R&I (MAS, n.d.), reflecting strong fiscal and external balance sheets, which are bolstered by substantial fiscal and external surpluses, high per capita income, a prudent macroeconomic policy

framework, and a favourable business environment (Fitch, 2024). The fiscal reserves are estimated to be between 200% and 300% of the GDP by Fitch (2024), ensuring that Singapore can cushion the impact of global tensions. This is appealing to investors as it shows that Singapore is a stable and low-risk environment for investments. The strong fiscal reserves and balance sheets also build investors' confidence in the government's capabilities to withstand economic pressures and remain resilient in the face of uncertainty.

Additionally, the Singapore government has consistently demonstrated a commitment to fiscal policies that promote fuel economic growth via investment programs and governmental expenditures. In addition to the various initiatives under Budget 2025, Singapore will establish the S$5 billion Equity Market Development Program towards the end of 2025 to boost the local equities market (Ngui, 2025). MAS will examine qualifying fund managers and program strategies in the coming months to actively manage and invest in a diverse range of Singaporean companies aside from index components, fuelling economic growth through an increase in job opportunities and employment. The program also includes a 20% tax rebate for primary listings to attract listings, along with a 10% tax rebate for new secondary listings with share issuance. The local equities market is anticipated to increase in trading volume and liquidity as more companies list, which will increase its value to both institutional and individual investors. This solidifies Singapore's standing as a major international financial hub, boosting business expansion, employment creation, and economic activity amidst global uncertainties. This move, along with the inexpensive valuations and high dividend yields, resulted in JP Morgan analysts setting the bull case target for the Straits Times Index (STI) of 4200, reflecting a 6% potential upside (Reuters, 2025). This validates the attractiveness of Singapore as an investment destination.

Part (b) - Appraisal of the Energy and Industrials Sector in Singapore

By 2050, the world's energy consumption is expected to increase by 11% in the case of Continued Momentum. The majority of this expansion will occur in emerging nations, where rising energy demand is a result of an expanding population (refer to Figure 1.2) (McKinsey, 2024).

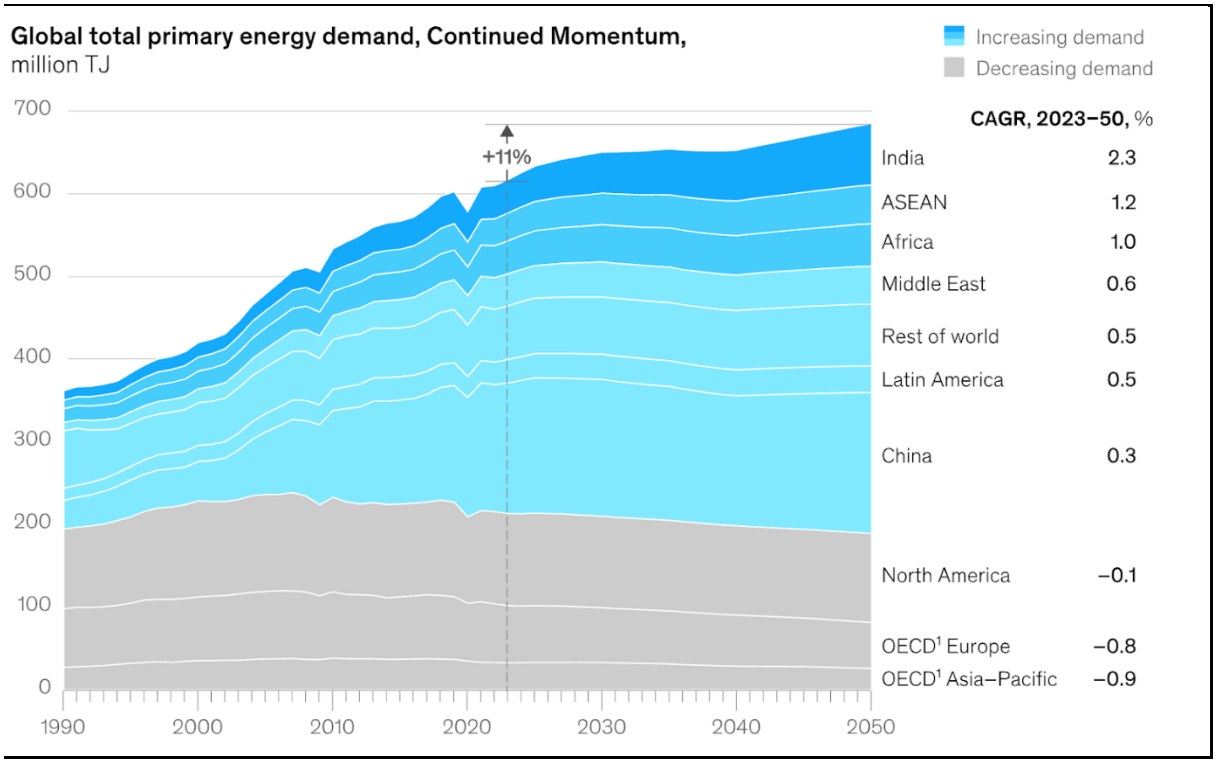

Figure 1.2 - Increase in demand for global energy (continued momentum)

The following summary table provides an analysis of the Energy and Industrials Sector using **Porter's Five Forces framework**.

1) Threat of New Entrants: **Moderate to High**

The Energy and Industrial sector is capital intensive, where it is expensive to invest and maintain the required infrastructure (e.g. power plants) and equipment. The sector contributes to an estimated one-third of the world's annual capital expenditure, and its capital intensity rate is more than twice that of sectors (BCG, 2023).

Licensing and permits are required to break into the Energy and Industrial sectors, as there are strict regulatory requirements. Moreover, there are policy frameworks to lower carbon emission rates. Singapore aims to achieve net-zero emissions by 2050 as per the Long-Term Low-Emissions Development Strategy (LEDS) (NCCS, n.d.). This means that there are relatively high barriers to entry which protect existing players in the sector. However, innovative new firms may be able to enter the sector, given that there is a large emphasis on the transition towards green energy transition.

2) Bargaining Power of Suppliers: **High**

The sector requires natural resources such as crude oil from suppliers (IEA, n.d.). As these resources are generally not available in Singapore, the suppliers have significant leverage on the firms, and subsequently, high bargaining power.

3) Bargaining Power of Buyers: **Low**

Manufacturing, construction and the government are some of the client bases the sector has access to. As the cost of switching energy infrastructure is high, the buyers are less likely to switch suppliers. This means that the bargaining power of buyers is relatively low.

4) Threat of Substitutes: **Moderate**

With the rise in sustainable energy, alternatives such as renewable energy (i.e. solar) are gaining traction. By 2030, solar and wind power are predicted to account for between 25% and 36% of the world's total power production (Crellin, 2024). By 2050, low-carbon energy is predicted to account for between 65% and 80% of the world's total power production (McKinsey, 2024).

However, economic pressure including inflation and high capital costs is slowing down the transition to cleaner energy (BCG, 2023). Moreover, key transition levers are not fully mature, scalable, or cost-effective to facilitate a complete transition towards green energy. Hence, fossil fuel demand will still account for around 40 - 60% of global demand in 2050 (McKinsey, 2024). This indicates a growing risk of substitution, hence the threat of substitutes is moderate.

5) Industry Rivalry: **High**

The sector includes key players such as Sembcorp, Keppel and ST Engineering. These firms operate in overlapping areas of businesses and hence have a similar client base. As such, there are close industry rivals that compete against and collaborate with each other to maintain their market share (Daga, 2022).

In response to extenuating global changes, Singapore has positioned itself to be a key player in the global energy supply market, with liquified natural gas (LNG) trade making up 50% of Singapore's GDP in 2022, an impressive feat since Singapore does not inherently have any LNG resources (Lee, 2023). Besides natural gas, Singapore is also home to some of the world's largest oil refining facilities, with large multinational corporations like ExxonMobil upgrading their refinery to meet rising regional demands in 2025 (Reuters, 2022), and Shell selling theirs off to a Swiss-Indonesian partnership with an outlook to shift the refinery's focus from transport fuels to chemicals to capitalise on the economy for chemicals (Yap, 2024). Furthermore, Singapore Green Plan 2030 envisions to align the country with zero-emission goals to achieve by 2050 while strengthening efforts towards achieving the UN's Sustainable Development Goals. The government has displayed their commitment through investments in various initiatives; the Energy Market Authority (EMA) has been conducting feasibility studies regarding Carbon Capture and Storage initiatives at selected power generation companies to find out if carbon capture can be utilised for energy production (Ng, 2024), while the Economic Development Board (EDB) has planned a $10 million investment into establishing the country as a clean energy hub by diversifying power sources through cross-border power grid partnerships and developing multi-energy solutions away from traditional fuels (Renewable Energy,n.d.).

With such initiatives and Singapore's strategic importance, the country is able to respond quickly to global energy trends and consistently present itself as a compelling investment destination. For instance, Singapore has positioned itself as a leader in regional LNG trade, capitalising on the demand for it, at home through bunkering facilities, and abroad as a clean energy alternative (SLNG, 2025). With global demand for LNG slated to increase around 60% in the coming 15 years driven by economic growth and emission reduction practices in the transportation industry, Singapore finds itself in a prime position to take advantage of its location and infrastructure to meet regional demands (Rashad & Chow, 2025). On top of that, aligning with the global shift towards net-zero emissions, the government has actively invested in solar energy and sought low-carbon energy alternatives to power homes. The government plans to utilise solar energy to power over 300,000 homes in Singapore annually by 2030 (NCCS, n.d.-b), while implementing the National Hydrogen Strategy for hydrogen to become an alternative electricity source handling the majority of the country's energy needs by 2050, effectively being a big contribution to the country's decarbonisation (MTI, n.d.).

Combining the support from the government's support along with Singapore's strategic location and the rising demand for sustainable energy alternatives, this makes Singapore's energy and industrial sector an attractive investment opportunity for investors.

Part (c) - Competition Analysis

The companies we have chosen are listed companies on the Singapore Exchange (SGX) under the industrials conglomerates sector and are involved in capital goods and business services. The three companies are Sembcorp Industries Ltd **(SGX: U96)**, Keppel Corporation Ltd **(SGX: BN4)** and Singapore Technologies Engineering Limited **(SGX: S63)**.

They are considered close competitors due to them operating in overlapping markets and providing solutions to similar client profiles, notably bidding for contracts with governments and private companies. Each company has been developing energy, infrastructure, and urban development solutions, and has been actively engaged in competition on multiple fronts, sometimes needing to collaborate to retain their market share. Sembcorp and Keppel decided to engage in a US$6.3 billion merger of their offshore and marine subsidiaries to consolidate their powers and positions in a bid to compete more effectively against their Chinese and South Korean rivals in 2022 (Daga, 2022), while ST Engineering, despite historically offering products related to the defense sector, has expanded their capabilities to encompass environmental engineering and sustainable infrastructure solutions, most notably testing a waste collection system in Tianjin worth over S$20 million in 2010, directly aligning themselves with some solutions offered by the other 2 (SBR, 2011). Combining their past competition along with similar financial standings in profit and revenue exemplifies their scopes and positions as major players in the industrials sector, as well as the dynamic nature of this industry where companies need to ideate and implement strategic business decisions to maintain market share.

To determine which of these companies is chosen, we assess their investment potential using appropriate financial ratios to determine the financial health of each company and make our comparison.

From the financial estimates of the three companies, in the last 12 months, Singapore Technologies Engineering Ltd (ST Engineering) has the highest total revenue of $11,275,659 followed by Keppel Ltd (Keppel) at $6,601,158 and Sembcorp Industries Ltd (Sembcorp) at $6,417,000. However, revenue alone would not accurately measure the operational performance as it does not factor costs, in terms of EBITDA, EBIT and Net Income, Sembcorp has outperformed its two peer competitors by a small margin (refer to Figure 1.3.1).

	B	C	D	L	P	T	X
7				Financial Estimates			
8				Total Revenue (S$000)	EBITDA (S$000)	EBIT (S$000)	Net Income (S$000)
9	Ticker		Company Name	LTM	LTM	LTM	LTM
10							
11	Target Company						
12							
13	SGX:U96		Sembcorp Industries Ltd (SGX:U96)	6,417,000	1,573,000	1,123,000	1,036,000
14							
15	Comparable Companies						
16							
17	SGX:BN4		Keppel Ltd. (SGX:BN4)	6,601,158	1,008,816	801,300	974,214
18	SGX:S63		ST Engineering Ltd (SGX:S63)	11,275,659	1,531,884	994,050	729,205
19							
20	**Minimum**			6,417,000.00	1,008,816.00	801,300.00	729,205.00
21	**Median**			6,601,158.00	1,531,884.00	994,050.00	974,214.00
22	**Average**			8,097,939.00	1,371,233.33	972,783.33	913,139.67
23	**Maximum**			11,275,659.00	1,573,000.00	1,123,000.00	1,036,000.00

Figure 1.3.1 Income Financials of Comparison Companies

Comparing the last twelve month multiples of the companies, Keppel performs better overall on the aspect of EV/Revenue, EV/EBITDA, EV/EBIT and P/E ratio, followed by ST Engineering followed by Sembcorp (refer to Figure 1.3.2). Further financial ratios were implemented for analysis and derived a myriad of results (refer to Figure 1.3.3). For a more in-depth analysis to decide on a company for further analysis, a comparison table with selected ratios has been done to exemplify our findings.

	B	C	D	AC	AG	AK	AO
7				Multiples			
8				EV/Revenue	EV/EBITDA	EV/EBIT	P/E
9	Ticker		Company Name	LTM	LTM	LTM	LTM
10							
11	Target Company						
12							
13	SGX:U96		Sembcorp Industries Ltd (SGX:U96)	3.01	12.28	17.20	10.68
14							
15	Comparable Companies						
16							
17	SGX:BN4		Keppel Ltd. (SGX:BN4)	3.36	22.00	27.70	13.03
18	SGX:S63		ST Engineering Ltd (SGX:S63)	2.22	16.32	25.15	27.72
19							
20	**Minimum**			2.22	12.28	17.20	10.68
21	**Median**			3.01	16.32	25.15	13.03
22	**Average**			2.86	16.87	23.35	17.15
23	**Maximum**			3.36	22.00	27.70	27.72

Figure 1.3.2 Valuation Ratios for Comparison Companies

	B	C	D	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM
7				Financial Ratios (Peers Comparison)									
8													
9	Ticker		Company Name	Gross %	EBITDA %	EBIT %	Net Income %	Total Asset Turnover (x)	ROA %	ROE %	Quick Ratio (x)	Current Ratio (x)	Book D/E
10													
11	Target Company												
12													
13	SGX:U96		Sembcorp Industries Ltd (SGX:U96)	23.45	0.25	0.18	0.16	0.38	4.17	19.85	0.97	1.16	166.11
14													
15	Comparable Companies												
16													
17	SGX:BN4		Keppel Ltd. (SGX:BN4)	28.10	0.15	0.12	0.15	0.24	1.84	7.72	0.92	1.39	105.66
18	SGX:S63		ST Engineering Ltd (SGX:S63)	19.28	0.14	0.09	0.06	0.71	3.93	25.57	0.55	0.89	197.28
19													
20	**Minimum**			19.28	0.14	0.09	0.06	0.24	1.84	7.72	0.55	0.89	105.66
21	**Median**			23.45	0.15	0.12	0.15	0.38	3.93	19.85	0.92	1.16	166.11
22	**Average**			23.61	0.18	0.13	0.12	0.44	3.31	17.71	0.81	1.15	156.35
23	**Maximum**			28.10	0.25	0.18	0.16	0.71	4.17	25.57	0.97	1.39	197.28

Figure 1.3.3 Financial Ratios for Comparison Companies

Table 2 - Comparison Table

Ratio Analysis

Financial Ratios

These ratios are used to analyse a company's financial health, performance, and its efficiency. The information they provide assists different stakeholders of the companies to make informed decisions.

1. Return on Equity

The Return on Equity (ROE) ratio provides insights into the firm's profitability, by measuring the amount of profit the firm earns with every $1 of equity. A higher value indicates that the firm is efficient in utilising the equity to generate profits.

Among the three firms, ST Engineering has the highest ROE at 25.57%, outperforming Sembcorp. This means that ST Engineering earned $0.2557 in profits for every $1 of shareholder equity, implying that the firm is highly profitable as the management is effective in generating returns using equity. This is followed by Sembcorp with an ROE of 19.85%, indicating that the company has earned $0.1985 in profits for every $1 of shareholder equity, signifying high profitability. Keppel has the lowest ROE at 7.72%, where only $0.0772 in profits for every $1 of shareholder equity, indicating low profitability.

2. Debt/Equity Ratio

The Debt to Equity (D/E) ratio measures the firm's financial leverage, by comparing the total debt against the total shareholders' equity. A higher value indicates that the firm relies on debt financing as compared to equity financing and vice versa.

Among the three firms, Sembcorp has the highest D/E ratio of 2.21. This means that the firm has $2.21 of debt for every $1 of equity. This shows that Sembcorp utilises a more aggressive debt financing structure, exposing the firm to financial risks amidst global uncertainties. This is followed by Keppel with a D/E ratio of 0.85, indicating that the company has $0.85 of debt for every $1 of equity. Out of the 3 firms, Keppel has the most balanced D/E ratio, indicating that it has a moderate approach towards financing. Lastly, ST Engineering has the lowest D/E ratio of 0.02, where there is $0.02 debt for every $1 of equity. ST Engineering's approach reflects conservative leverage, with minimal to no reliance on debt. This may limit the firm's growth potential, as even their equity of $1,984,943,000 is considerably lower than that of Sembcorp at $5,659,000,000 and Keppel at $11,425,661,000.

3. Quick Ratio

The Quick Ratio measures the firm's liquidity by quantifying its capacity to fulfil short-term liabilities with liquid assets. A higher value indicates that the firm is more liquid in generating cash.

Sembcorp has a near-perfect quick ratio of 0.97 which is considerably close to the ideal ratio of 1. This shows that Sembcorp has $0.97 of liquid assets for every $1 of current liabilities, suggesting that it has nearly sufficient liquidity to fully meet short-term liabilities. Similarly, Keppel has a quick ratio of 0.92, which is slightly weaker than Sembcorp. This means that Keppel has $0.92 of liquid assets for every $1 of current liabilities, indicating stable liquidity. ST Engineering, on the other hand, has a significantly lower quick ratio than the other two firms. Its ratio of 0.55 shows low liquidity, with only $0.55 of liquid assets for every $1 of current liabilities. This means that ST Engineering is exposed to the risk of meeting short-term liabilities without external financing.

4. Net Profit Margin

The Net Profit Margin measures the firm's profitability and management of expenses, by computing the percentage of the revenue that is generated into net income. A higher value indicates that the firm can generate a higher net profit and manage expenses effectively.

Sembcorp has the highest Net Profit Margin at 16.14%, showing that the firm has generated $0.1614 in net profit for every $1 in revenue. This indicates that Sembcorp performed the best in terms of profitability and cost control. The ratio suggests that Sembcorp has efficient operations and expense management, which enabled the firm to retain more profits. This is followed closely by Keppel with a Net Profit Margin of 14.76%, where $0.1476 in net profit was earned for every $1 in revenue. This means that Keppel also has good profitability even though its performance is not as strong as Sembcorp's. On the other hand, ST Engineering has the weakest Net Profit Margin, at 6.47%, illustrating that only $0.0647 in net profit was earned for every $1 in revenue. The significantly lower margin suggests that ST Engineering may have deficiencies in its cost management.

<u>Valuation Ratios:</u>

These ratios can be analysed to assess a company's financial worth and if it poses a good investment opportunity. By comparing different companies within the same industry, it provides investors with a brief understanding of the market sentiment, determining if the stock is worth investing in.

1. Price-Earnings (P/E) Ratio

This ratio measures the stock market's expectation for the company's future growth with regards to its earnings, dividends, and future opportunities. A higher value indicates that investors can expect higher future growth.

Among the 3 companies, ST Engineering has the highest P/E ratio at 27.72. This indicates to investors that they can be expectant of strong growth by the company in the future. On the other hand, Keppel's currently stands at 13.03 and Sembcorp's at 10.68. This could be an indication that their stocks are relatively undervalued compared to that of ST Engineering's, carrying a lower price tag which could be attractive to investors who believe that they can get the stock for a bargain. There would also be potential for strong stock appreciation, making them a growth-at-a-reasonable-price (GARP) investment opportunity.

2. Enterprise Value to EBITDA

This ratio indicates the company's value in relation to its operating profitability. This is useful for valuing large companies with complex capital structures, which is essential here

given that all 3 companies are large global conglomerates. A lower value would suggest that the company's stock is undervalued.

Among the 3 companies, Keppel's EV/EBITDA is the highest at 22.00, as compared to Sembcorp's at 12.28 and ST Engineering's at 16.32, both which are considerably lower. This could signal to investors that the 2 latter companies pose a better investment opportunity than that of the former, as their stock prices are undervalued, showing potential for future appreciation.

3. Enterprise Value to Revenue

This ratio shows how much investors are willing to pay per dollar of revenue. A lower value suggests the stock being undervalued, while a high value indicates optimism on the company's stock's growth.

Among the 3 companies, Keppel's EV/Revenue is the highest at 3.36, followed closely by Sembcorp's at 3.01 and then ST Engineering's at 2.22. Given that the global average for EV/Revenue ranges from 1 to 3, this would indicate that Keppel and Sembcorp's EV/Revenue is outperforming, which could mean either the stock is being overvalued or that there are strong growth expectations for both companies. This tells investors that both companies are good investment choices, but they should conduct their due diligence on each company's revenue growth to determine its legitimacy before putting down an investment.

Company of Choice for Further Analysis

After meticulous evaluation of the 3 firms, the team decided to select Sembcorp Industries Ltd **(SGX: U96)** for further investment analysis in Question 2.

Although ST Engineering has strong ROE and P/E ratio performance, its lower Net Profit Margin and D/E ratios suggest lower growth potential. Keppel had stronger EV/Revenue and EV/EBITA metrics but fell short in its ROE and Net Profit Margin, indicating a potential shortfall in operations.

Sembcorp's diverse business operations across different areas such as energy, infrastructure, and urban development position it as a major renewables player in Asia. Sembcorp has proven its profitability capacity with its commendable ROE of 19.85% and Net Profit Margin of 16.14%. The firm's sound quick ratio of 0.97 also provides security against short-term financial risks. The high Debt to Equity ratio of 2.21 reflects the high leverage of debt to finance growth, given that it operates in capital-intensive businesses. Although Sembcorp is vulnerable to financial risks in periods of decline in the economy, its profitability and liquidity can bolster the impacts of debts.

Furthermore, Sembcorp's stock is attractively priced relative to its earnings with a respectable PE ratio of 10.68. As it is lower than Keppel's PE ratio of 13.03, it shows that Sembcorp is potentially undervalued. Sembcorp's EV/Revenue of 3.01 is above the global average, but it is lower than Keppel's Keppel's EV/Revenue of 3.36, indicating potential growth expectations. Similarly, Sembcorp's EV/EBITA of 12.28 is also lower than that of

Keppel at 22, which reinforces the narrative that Sembcorp has growth potential as compared to its industry peers.

In conclusion, Sembcorp has a balanced performance with the given valuation and financial ratios with strong profitability and liquidity. The firm is well-positioned to leverage the growing traction for green energy due to its solid position in the renewables industry and its strategic focus on sustainability and expansion. As part of its 2023 - 2028 strategic plan, Sembcorp is also investing $10.5 billion in renewable energy to amplify its growth in the renewables space (Tay, 2024). This makes Sembcorp a suitable growth-centric investment option, positioning it as a growth-centred investment option amidst global uncertainties.

Question 2

Part (a) – Free Cash Flow to Firm and Free Cash Flow to Equity

Revenue breakdown by geography		
Country	**Revenue (000)**	**% of Total Revenue**
Singapore	4,754,000	74.08%
UK	561,000	8.74%
China	383,000	5.97%
India	366,000	5.70%
Rest of Asia	287,000	4.47%
Middle East	48,000	0.75%
Other Countries	18,000	0.28%
Total	**6,417,000**	**100.00%**

Table 2 – Revenue Breakdown by Geography

i)	Operating Cash Flow	2020A	2021A	2022A	2023A	2024A	Source:
	OCF = EBIT(1 – Tax Rate) + Depreciation						
	EBIT (Operating Income)	569	540	897	1,177	1,106	*Income Statement*
	EBIT(1-Tax Rate)	472	448	745	977	918	*Assume Singapore corporate tax rate of 17%*
	Add: Depreciation	455	333	385	466	459	*Cashflow Statement; Operating Activities*
	OCF	**927**	**781**	**1,130**	**1,443**	**1,377**	***=SUM(G7:G8)***
iii)	**Changes in NWC**	**2020A**	**2021A**	**2022A**	**2023A**	**2024A**	
	Changes in working capital:						*Cashflow Statement; Operating Activities*
	Inventories	(50)	(28)	(51)	2	(5)	
	Account Receivables	(214)	(483)	276	169	(119)	
	Account Payables	(301)	498	(212)	(230)	140	
	Unearned Revenue	118	(17)	13	43	25	
	Other Net Operating Assets	(5)	-	(2)	-	-	
	Total changes working capital	**(452)**	**(30)**	**24**	**(16)**	**41**	***=SUM(G13:G17)***
							z
	Free Cash Flow to Firm	**2020A**	**2021A**	**2022A**	**2023A**	**2024A**	
	FCFF = OCF – NCS – ΔNWC						
i)	Operating Cash Flow (OCF)	927	781	1,130	1,443	1,377	
ii)	Less: Net Capital Spending (NCS)	(333)	(290)	(615)	(836)	(1,608)	*Cashflow Statement (Capital Expenditure)*
iii)	Less: Changes in NWC (ΔNWC)	(452)	(30)	24	(16)	41	*Cashflow Statement; Operating Activities*
	FCFF	**142**	**461**	**539**	**591**	**(190)**	***=SUM(G23:G25)***

Figure 2.1.1 - Calculation of free cash flow to firm (FCFF)

Free Cash Flow to Firm (FCFF)

$$FCFF = OCF - NCS - \Delta NWC$$

Operating Cash Flow (OCF):

1. OCF = EBIT(1 – Tax Rate) + Depreciation
2. OCF = NI + Depre + Int Exp(1 – Tax Rate)
3. OCF = EBITDA(1 – Tax Rate) + Depreciation(Tax Rate)

To derive OCF in Figure 2.1.1, there are 3 possible formulas above, of which we utilised the first. Since a majority 74% of the company's revenue comes from Singapore, a corporate tax rate of 17% is used (Table 2).

Net capital spending (NCS) included company purchases of property, plant, and equipment (PPE), investment properties and intangibles.

As for the change in working capital (ΔNWC), only the line items in the figure above are included as these were identified in the cashflow statement (operating activities) by the company to be their specific working capital in CFS. Hence, that was used for better representation of the company's ΔNWC instead of using the net value of the all-inclusive current assets and liabilities (CA-CL).

Free Cash Flow to Equity (FCFE)

	Free Cash Flow to Equity	2020A	2021A	2022A	2023A	2024A	
	FCFE = NI + Depreciation + Net Borowings – NCS – ΔNWC						
i)	NI	(997)	279	848	942	1,011	*Income Statement*
ii)	Add: Depreciation	455	333	385	466	459	*Cashflow Statement; Operating Activities*
iii)	Add: Net Borrowings	862	(364)	287	(437)	924	
iv)	Less: NCS (Capex)	333	290	615	836	1,608	*Cashflow Statement (Capital Expenditure)*
v)	Less: ΔNWC	(452)	(30)	24	(16)	41	*Cashflow Statement; Operating Activities*
	FCFE	**439**	**(12)**	**881**	**151**	**745**	***=O13+O14+O15-O16-O17***

Figure 2.1.2 - Calculation of free cash flow to equity (FCFE)

These are the three possible formulas to obtain Free Cashflow to Equity (FCFE):

1. FCFE = NI + Depreciation + Net Borrowings – NCS – ΔNWC
2. FCFE = FCFF – Interest Expense (1 – Tax Rate) + net borrowings
3. FCFE = NI – [(1-DR) (NCS – Depreciation)] – [(1-DR) ΔNWC]

For Sembcorp we used Method (1) to calculate our FCFE value. Method (1) is the most straightforward method as it uses the net income, which is mostly readily available from the financial statement, this method focuses directly on the cash flow available to equity holders, it also accounts for non-cash expenses like depreciation as well as net working capital changes for cash flow.

Method (2) uses Free Cash Flow to Firm (FCFF) as a starting point which is a broad measure of cash flow available to both debt and equity holders. This reflects the cash flow available to equity holders after debt obligations are met, however, it can be complex as it requires FCFF for FCFE to be calculated and is sensitive to changes in the interest rates and debt as Sembcorp possesses a significant amount of debt to equity.

Method (3) is not used as Sembcorp does not explicitly have a target debt ratio (DR) as the debt level fluctuates over the period due to reinvestments.

Part (b) – Projected Growth Rates of Company (High & Normal)

Sembcorp Industries (SCI) is likely to experience high revenue growth over the next five years, driven by key catalysts in its strategic plans for its renewable energy and urban development sectors. Since 2023, SCI has been accelerating the alignment of its sustainable portfolio, with the global energy transition goals of Singapore, heading towards a low-carbon future by 2028.

A key growth strategy for SCI is its aggressive expansion in renewable energy. The company expanded its gross renewables capacity by four times to about 12.0GW, which now accounts for 61% of its energy portfolio. Furthermore, SCI is on track for its interim target of 10GW by 2025 and aims to double this figure to 25GW by the year 2028. This positions SCI as a leading renewables player in Asia and stands among the top five electricity suppliers in Singapore, its total gross power capacity stands at 16.7GW (Sembcorp Industries, n.d.).

SCI's leadership in Singapore's renewable energy landscape is further supported by its success in securing solar lease tenders to date as they reinforce SCI's dominant position in the market. Other catalysts such as their involvement in short-term mobile solar projects at PSA's Keppel Terminal generate 22,500 megawatt hours of electricity annually—enough to power over 5,000 four-room HDB flats. With electricity demand projected to grow at a CAGR of 4-6% by 2028, SCI is well-positioned to capitalise on this trend, particularly through its expertise in solar leasing and generation (Sembcorp, 2023).

Looking ahead, SCI is preparing for emerging opportunities in low-carbon energy, such as hydrogen production and carbon management, anticipating their commercial and technological maturity. This addresses customer needs but also balances energy security, accessibility, and affordability. By combining its strong track record, strategic investments, and alignment with Singapore's sustainability goals, SCI is solidifying its role as a key driver of the region's energy transition and a high-growth player in the renewable energy space.

Sembcorp is in the energy and infrastructure industry with four revenue segments. We start by calculating the market size for each revenue segment:

1) **Gas and Related Services (GRS)**
   A statistic published by Statista claims that Singapore's GRS industry has a market size of $2.8 billion in revenue that is up for grabs, with a CAGR of 4% (Romero, 2024).

2) **Renewables**
   Another statistic published by Statista predicts the market demand in Singapore for renewables is set to rise to $1.28 billion by 2025, with a CAGR of 3.19% (Statista, 2025).

3) **Integrated Urban Solution (IUS)**
   A GlobalData report published in December 2024 forecasts a market size of a whopping $26 billion in Singapore's IUS industry, with a CAGR of 4.4% (GlobalData, 2024).

4) **Decarbonisation Solutions (DS)**
   Finally, Fortune Business Insights anticipates that Singapore's DS market size will be valued at 1.39 billion by the end of 2025, with a CAGR of 4.4% (FBI, 2025).

In total, this means Sembcorp's revenue segments are all involved in exciting markets that have a good addressable size, totalling $31.47 billion in total, with a median CAGR of 4.2%.

Statista published a reported market share of energy companies in Singapore, Figure 2.2 below.

Top Electricity Companies in Singapore 2024

Here are the top 7 **energy companies in Singapore**:

Company	Market Share
Tuas Power Generation Pte. Ltd.	20.20%
Senoko Energy Pte. Ltd.	17.60%
Keppel Merlimau Cogen Pte. Ltd.	13.90%
YTL PowerSeraya Pte. Ltd.	13.10%
SembCorp Cogen Pte. Ltd.	11.80%
PacificLight Power Pte. Ltd.	9.80%
Tuaspring Pte. Ltd.	3.10%
Others	10.70%

Source: Statista.com

Figure 2.2 Market Share of Energy Companies in Singapore

In November 2024, Sembcorp announced that it has completed an acquisition of 30% stake in Senoko Energy (Sembcorp, 2024). This means that currently, Sembcorps market share is 11.80% plus 30% of Senoko Energy amounting, raising Sembcorp's total market share in energy and infrastructure to 17.08% (30%*17.60%). Increasing its ranking from the top 5th to top 2nd in Singapore, 2024. Their market share of 17% makes Sembcorp Industries a dominant player where it is able to maintain its status in the market for the next 5 years or so.

To determine Sembcorp's growth rate, the following steps have been considered:

1) We expect the energy and infrastructure market to maintain its 4.2% CAGR in line with the Singapore 2030 Green Plan in the next 5 years. As such the market size is expected to grow from $31.47 billion in 2025 to $38.66 billion by the end of 2029.

2) Expecting Sembcorp to maintain its market share of 17% for the next 5 years, the expected revenue of Sembcorp in 2029 is 18% of $38.66 billion, amounting to $6.96 billion.

3) Sembcorp's annual growth rate is calculated to be 6.47% in the next 5 years, a compound annual growth rate CAGR of 6.47% will allow Sembcorp's revenue to grow from $4.8 billion today to reach the target $6.96 billion in the next 5 years.

4) After 5 years, the growth rate is assumed to return to a stable rate of 2.5%, slightly lower than the upper limit of GDP growth of Singapore's economy (1 to 3%).

Part (c) – Discounted Cash Flow Valuation Method

Given the choice between Dividends, Residual Income, FCFE or FCFF for our DCF model, **we have decided to use FCFF**. We used the following elimination methods:

1) Dividends
   a) During the stages of high growth, we expect Sembcorp not to pay out all its free cash flows as dividends as reinvestment is needed. However, in order to appease shareholders, we also expect Sembcorp to maintain moderate levels of dividend payments as it has even in times of high investment needs. This is due to the expectations of its "blue chip" reputation.

   b) This makes it **hard to estimate Sembcorp's dividend policy** in the next 5 years of high growth. Additionally, assuming dividends payout are forced to remain stable, the dividends will not be a good indicator to estimate market expectation.

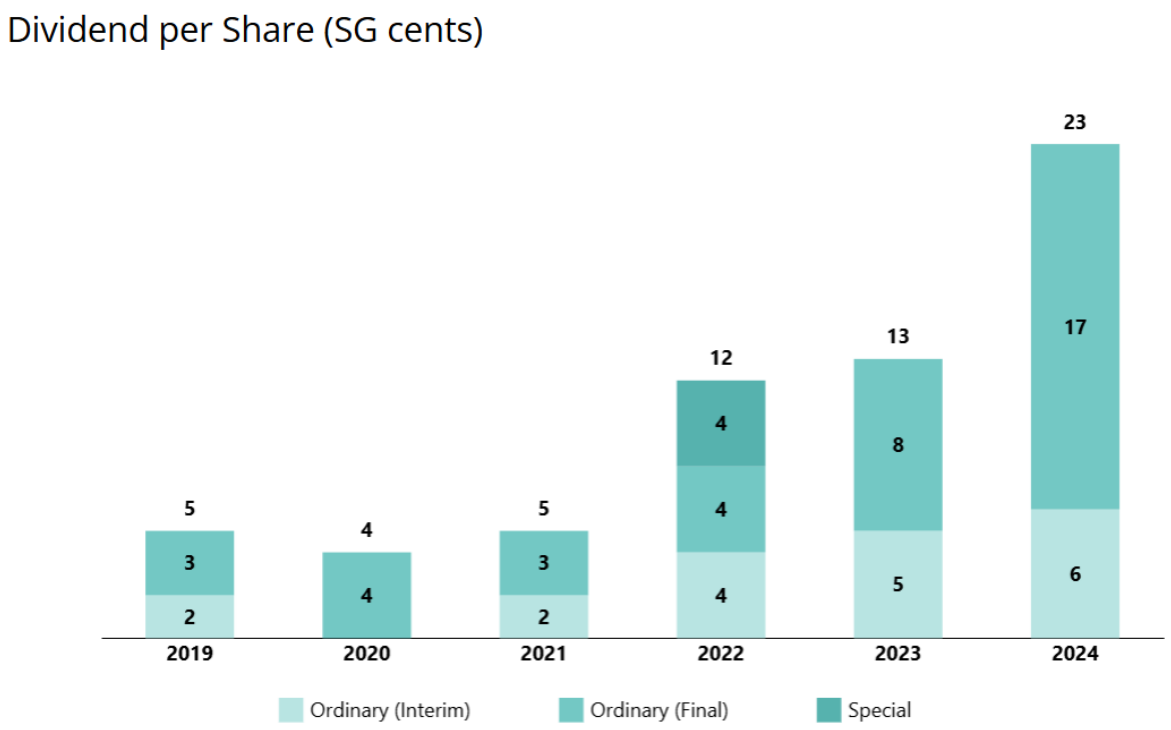

Figure 2.3.1 Sembcorp Dividend History (2019 - 2024)

2) Residual Income (RI)
   a) In Sembcorp's case the **clean surplus relationship does not hold**.

   b) Numbers are expressed in thousands, sourced from S&P Capital IQ:
      i) Beg. Book value of equity (2023): 4,872,000
      ii) End. Book Value of Equity (2024): 5,659,000
      iii) Dividends Paid (2024): 250,000
      iv) Net Income (2024): 1,011,000

c) Change in Book Value of Equity = 787,000

d) Change in NI minus Dividends (Retained Earnings) = 761,000

3) FCFE

a) We expect Sembcorp's capital structure to change over time due to the industry's high capital and project financing characteristics. Sembcorp also engages in large-scale infrastructure projects, often involving complex financing arrangements.

b) Additionally, Sembcorp is a complex company with diverse revenue segments, products and services. The FCFE model is better for companies with a single division or product due to its weaker ability to capture a holistic view of different forms of debt financing. Hence, the **FCFE will not be used.**

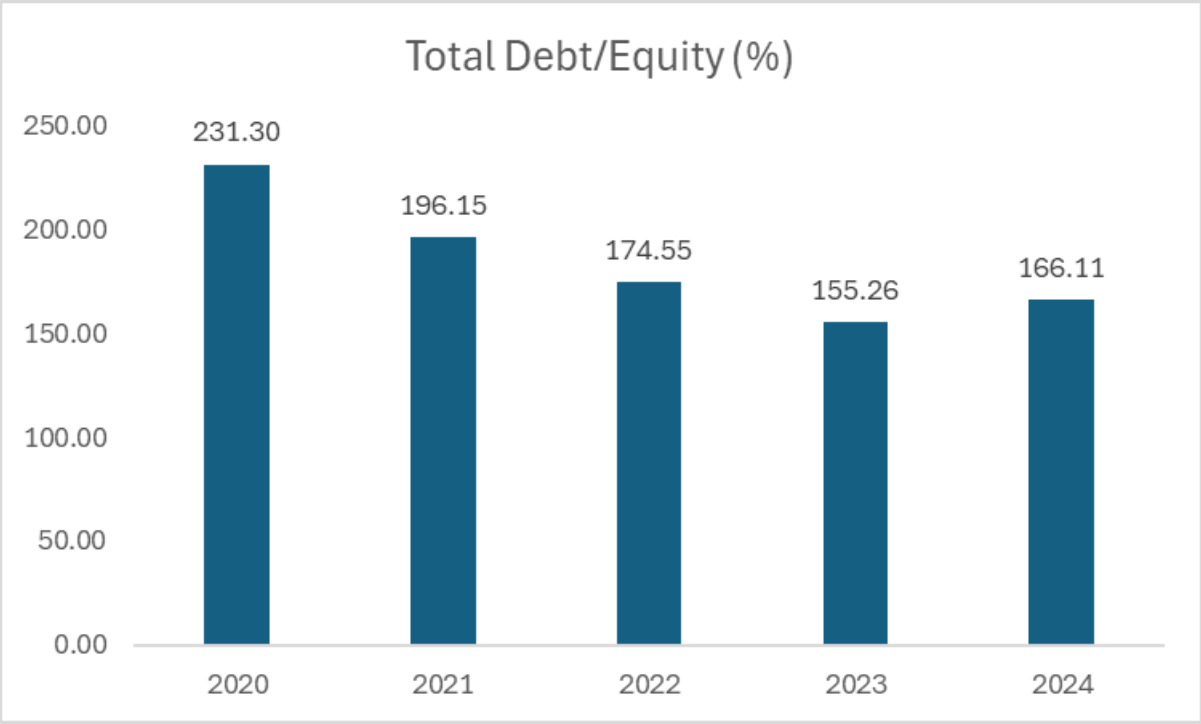

Figure 2.3.2 Sembcorp Debt-to-Equity Percentage

4) FCFF

a) FCFF values the entire firm, regardless of its capital structure; this is important because Sembcop's capital structure is expected to change. Sembcorp operates in capital-intensive sectors like utilities and infrastructure, where long-term projects and significant capital expenditures are common. FCFF is well-suited for valuing companies in these sectors because it focuses on the operational cash flows generated by these long-term assets, which are critical for valuation.

b) FCFF captures cash flows generated by projects before considering the impact of debt financing, providing a clear picture of the operational performance of the underlying assets.

To conclude, we have decided that **FCFF is the most appropriate.**

Given the choice of a single, three-stage or two-stage growth model. We have decided to **use a two-stage model**. We used the following elimination methods:

1) Single Stage FCFF Model
   a) We are assuming that our FCFF will not grow at a constant rate g so a single stage is **not suitable.**

2) Three Stage FCFF Model
   a) While a three-stage FCFF model aims to capture nuanced growth patterns, the added complexity **introduces significant estimation risk**. Each additional stage requires more assumptions about growth rates, discount rates, and the timing of transitions. This complexity can lead to a higher likelihood of errors and a reduction in the model's overall reliability. The potential for skewed results outweighs the perceived benefits of finer-grained predictions.

   b) Accurate data for long-term forecasts, especially for multiple distinct growth phases, is often limited. Relying on overly speculative projections for extended periods increases the **risk of inaccurate valuations**. It's preferable to use a model that relies on more readily available and reliable data.

   c) For a company like Sembcorp, operating in the utilities and infrastructure sectors, long periods of extremely high growth are less common. A two-stage model, which captures an initial growth phase followed by a stable growth period, is generally sufficient to reflect the company's expected trajectory. The added complexity of a third stage is **unlikely to provide** a significant improvement in accuracy.

3) Two Stage FCFF Model
   a) The two-stage FCFF model provides a reasonable balance between capturing the company's growth trajectory and maintaining simplicity. It avoids the excessive **complexity of a three-stage model** while still accounting for distinct growth phases.

   b) A two-stage model **aligns with the typical lifecycle** of many companies, including Sembcorp. It captures an initial period of higher growth driven by strategic initiatives, project expansions, or market penetration, followed by a transition to a more stable, long-term growth phase.

c) The two-stage model requires forecasting cash flows for a manageable period during the initial growth stage, followed by a stable growth assumption. This reduces the reliance on overly speculative long-term projections, **enhancing the model's reliability**.

d) The two-stage FCFF model is **widely accepted and understood** by analysts and investors. This facilitates communication and comparison with other valuation analyses. Its simplicity enhances transparency and makes it easier to explain the valuation process.

To conclude, we will implement a two-stage FCFF model with a **high growth rate of 7.70%** in the first 5 years and **a stable growth rate of 2.5%** slightly below Singapore's GDP after 5 years.

Part (d) – Implementation of the Chosen Valuation Model

We begin by identifying and justifying the growth rates of each line item in our FCFF calculation.

	Year 1 - 5 (Growth)	After Year 5 (Growth)
Revenue	6.47%	2.5%
Operating Margin	Grows from 13% to 16%	16%
Tax Rate	0% (remains at 17%)	
Depreciation & Amortization	50%	80%
Change in NWC	1.24%	0.6%
Sales to Capital Ratio	0.44	0.83
WACC	6.52%	5.60%

1) Revenue

The initial high-growth rate of 6.47% from 2026 to 2029 is justified by Sembcorp's increasing market share within the expanding Energy and Infrastructure sectors. This growth rate is supported by industry reports projecting (refer to Q2B) for these markets, in which Sembcorp actively participates.

The Ministry of Trade & Industry (MTI) Singapore forecast the GDP growth rate of 2025 to be between 1 to 3%, while most economists forecasted a GDP between 2% to 3% (Tan, 2025). The global tension among the US and China trade tariffs are the main contributors to the economic uncertainty. However, Singapore's political stability and social cohesion does provide certain optimism in our economy (Yee, 2024). We will be using a fair but conservative view of our sustainable growth rate of 2.5% (IMF, n.d.).

2) Operating Margin

The initial fluctuation in operating margins, decreasing from 16.29% to 13% and then gradually increasing to 16%, reflects the impact of initial costs associated with new projects, changes in input costs, or strategic investments. For example, Sembcorp is in the process of completing a $900 million multi-utilities development on Jurong Island that is expected to be functional by 2026 (Devi, 2024). This is one such project amidst many others, Sembcorp has been awarded the assembly of a short-term mobile solar farm (Soh, 2025) and the importation of green energy in partnership with Sarawak Energy (See, 2025). The increase back to 16% reflects the projects becoming fully operational and efficient.

The stabilization of the operating margin at 16% from 2028 onwards represents the company's mature operational efficiency and long-term profitability within its core business segments.

3) Tax Rate

The consistent tax rate of 17% is based on the current corporate tax rate in Singapore and reflects the company's expected effective tax rate, considering applicable tax incentives and regulations. Hence, it can also be considered the **marginal tax rate.**

4) Depreciation and Amortisation

The D&A is calculated as a percentage of reinvestment, reflecting the depreciation of newly acquired assets. The initial D&A growth rate of 50% are used to reflect the large amount of upfront capital expenditures on new projects.

The 80% used in the stable growth period reflects a more mature company, where the amount of depreciation is closer to the amount of reinvestment.

5) Change in NWC

The change in Net Working Capital (NWC) is calculated as a percentage of the change in revenue, reflecting the company's working capital needs as it expands its operations. During the high-growth period (2025-2029), a rate of 1.24% is used. This rate is derived from an analysis of smaller, more agile competitors within the Energy and Infrastructure sectors in which Sembcorp operates. These competitors, while smaller, exhibit similar operational characteristics and working capital management practices as Sembcorp. One such company is Union Gas Holdings Limited with a net working capital increase of 1.24% of its revenue in 2024 (Figure 2.4.1).

Figure 2.4.1 Growth in Change in Net Working Capital

The 0.6% used in the stable growth period, reflects the stable working capital needs of a mature company. The 0.6% was derived from Sembcorp's current % of NWC to revenue.

The large increase in the sales to capital ratio, means that the company is much more efficient, and therefore the percentage of NWC to revenue is lower.

6) Sales to Capital Ratio

The initial sales-to-capital ratio of 0.44 reflects Sembcorp's current asset utilization and capital efficiency based on historical financial data. This ratio is characteristic of companies operating in capital-intensive sectors like utilities and infrastructure, where significant investments in long-lived assets are necessary.

Furthermore, this ratio suggests that Sembcorp is in a phase where it is investing heavily in growth initiatives, including marketing efforts to expand its customer base and hiring skilled workers to support its operational capabilities. These investments,

while essential for long-term growth, may initially result in a lower sales-to-capital ratio as the company's assets are not yet fully optimized for revenue generation.

The significant increase in the sales-to-capital ratio to 0.83 from 2029 onwards is primarily attributed to a combination of operational efficiency improvements and strategic asset divestitures. Sembcorp has successfully implemented advanced digital optimization tools across its power and water treatment plants, leading to increased asset utilization and reduced operational costs. Additionally, the company has strategically divested non-core assets, allowing it to focus on its core, high-performing infrastructure projects. These initiatives have collectively resulted in a substantial increase in revenue generation relative to the company's capital base.

The 0.83 sales-to-capital ratio was derived from a median of Keppel and ST Engineering's sales-to-capital ratios for the year 2024 (Figure 2.4.2). This is a good representation of an industry ratio as the company returns to normal growth.

Figure 2.4.2 Peers Sales to Capital Ratio

7) WACC

During the high-growth stage, Sembcorp is undertaking significant expansion projects, entering new markets, and developing new technologies. These activities increase the company's operational complexity and introduce greater uncertainty regarding future cash flows. As a result, the business risk is higher during this period, justifying a higher WACC of 6.52%. The 6.52% was derived from an increase in Sembcorp's cost of debt to 7%.

The WACC in the normal growth is the company's current WACC of 5.76%. The WACC was derived using a blended revenue approach by geography (Appendix B & C).

	Year 1 - 5 (Growth)	After Year 5 (Growth)
Revenue	6.47%	2.50%
Operating Margin	Grows from 13% to 16%	16%
Tax Rate	0% (remains at 17%)	
Depreciation & Amortization	50%	80%
Change in NWC	1.24%	0.60%
Sales to Capital Ratio	0.44	0.83

	High G	Normal G
WACC	6.65%	5.76%

		High Growth					Terminal				
	Base Year (2024)	2025A	2026A	2027A	2028A	2029A	2030A	2031A	2032A	2033A	2034A
Revenue	4,802	5,113	5,444	5,796	6,171	6,571	6,735	6,903	7,076	7,253	7,434
Operating Margin	16%	13%	14%	15%	16%	16%	16%	16%	16%	16%	16%
Tax Rate	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%	17%
EBIT (1 - t)	638	552	633	722	820	873	894	917	940	963	987
Depreciation & Amortization	-	353	376	400	226	241	158	162	166	170	175
Change in NWC	-	63	68	72	77	81	40	41	42	44	45
Reinvestment	-	706	752	801	452	481	198	203	208	213	218
FCFF	-	**135**	**189**	**249**	**517**	**551**	**814**	**835**	**856**	**877**	**899**
PV of FCFF	-	**127**	**166**	**206**	**400**	**399**					

Sum of PV of FCFF	1,297	=SUM(D21:H21)
Terminal Value (2030)	24,949	=I19/(H3-D3)
PV of Terminal Value	17,824	=PV(H3,I20,,-C24)
Enterprise Value	19,122	=C23+C25
Equity Value	19,112	=C26-'WACC (normal growth)'!B17
Value per Share	**10.74**	=C27/'WACC (normal growth)'!B16
Stock Price (19 March 2025)	$6.20	
Upside/(Downside)	73.22%	

Figure 2.4.3 FCFF Calculation with 73.22% Projected Upside

Appendix

Appendix A - Sembcorp Industries Ltd Financial Statements

Sembcorp Industries Ltd (SGX: U96)	Historical				
For Fiscal Year Ended 31 Dec, SGD million	2020A	2021A	2022A	2023A	2024A
Income Statement					
Revenue	**5,447**	**6,408**	**7,825**	**7,042**	**6,417**
Cost of Sales	(4,660)	(5,589)	(6,598)	(5,469)	(4,912)
Gross Profit	**787**	**819**	**1,227**	**1,573**	**1,505**
General and administrative expenses	(344)	(393)	(499)	(432)	(476)
Other operating income, net	126	114	169	36	77
Operating Income	**569**	**540**	**897**	**1,177**	**1,106**
Non-operating income	49	21	9	148	186
Non-operating expenses	(176)	(218)	(16)	(7)	(13)
Finance income	35	21	37	57	27
Finance cost	(499)	(296)	(310)	(409)	(372)
Share of results of associates and joint ventures, net of tax	233	206	248	264	317
Profit/(Loss) before tax	**211**	**274**	**865**	**1,230**	**1,251**
Tax expense	(32)	(123)	(138)	(182)	(206)
Tax rate	*15%*	*45%*	*16%*	*15%*	*16%*
Profit/(Loss) from continuing operations	**179**	**151**	**727**	**1,048**	**1,045**
Profit from discontinued operations, net of tax	(330)	149	144	(78)	(9)
Profit/(Loss) on the distribution	(970)	-	-	-	-
Profit/(Loss) for the period	**(1,121)**	**300**	**871**	**970**	**1,036**
Attributable to:					
Net Income (attributable to owners of the company)	**(997)**	**279**	**848**	**942**	**1,011**
Non-controlling interest	(124)	21	23	28	25

Balance Sheet					
ASSETS					
Current Assets					
Inventories	196	222	137	135	135
Trade and Other Receivables	1,571	1,986	1,564	1,674	1,812
Contract Asset	15	28	29	15	37
Other investments and derivative assets	-	-	-	114	114
Contract Costs	1	1	3	1	1
Cash and Cash Equivalents	1,032	1,344	1,254	767	871
Other Financial Assets	159	352	89	-	-
Asset Held for Sale	30	-	3,432	-	392
Total Current Assets	**3,004**	**3,933**	**6,508**	**2,706**	**3,362**
Non-Current Assets					
Property, plant and equipment	7,204	7,094	5,305	6,465	8,304
Investment Properties	135	138	133	153	207
Interests in Associates and Joint Ventures	1,588	1,600	2,287	2,396	2,740
Intangible Assets	348	390	697	952	977
DPN receivable	-	-	-	1,816	1,581
Trade and Other Receivables	995	982	855	811	802
Contract Costs	1	1	-	-	-
Other Financial Assets	250	219	-	-	-
Other investments and derivative assets	-	-	183	132	136
Deferred Tax Assets	37	38	52	66	69
Total Non-Current Assets	**10,558**	**10,462**	**9,512**	**12,791**	**14,816**
Total Assets	**13,562**	**14,395**	**16,020**	**15,497**	**18,178**
LIABILITIES & EQUITY					
Current Liabilities					
Trade and Other Payables	1,159	1,708	1,715	1,630	1,585
Contract Liabilities	141	121	139	171	197
Lease Liabilities	11	14	17	18	27
Derivative Liabilities	-	-	99	63	36
Current Tax Payable	157	181	219	236	182
Loans and Borrowings	593	754	1,096	1,281	671
Provisions	26	40	42	77	65
Liabilities Held for Sale	-	-	1,494	-	148
Other Financial Liabilities	40	87	-	-	-
Total Current Liabilities	**2,127**	**2,905**	**4,821**	**3,476**	**2,911**
Non-Current Liabilities					
Loans and Borrowings	7,135	6,637	5,974	5,973	8,000
Lease Liabilities	215	244	270	292	702
Derivative Liabilities	-	-	23	20	30
Contract Liabilities	71	74	69	80	79
Deferred Taxation	294	392	492	598	629
Provisions	38	64	62	65	69
Other Financial Liabilities	98	56	-	-	-
Other Long-term Payables	108	105	93	121	99
Total Non-Current Liabilities	**7,959**	**7,572**	**6,983**	**7,149**	**9,608**
Total Liabilities	**10,086**	**10,477**	**11,804**	**10,625**	**12,519**
Shareholders' Equity					
Common Stock - Par Value	566	566	566	566	566
Treasury Stock - Common	(11)	(15)	(31)	(40)	-
Other Reserves	(369)	(133)	(608)	(664)	(676)
Revenue Reserve	3,153	3,349	4,050	4,726	5,471
Total	**3,339**	**3,767**	**3,977**	**4,588**	**5,361**
Minority Interest	137	151	239	284	298
Total Shareholders Equity	**3,476**	**3,918**	**4,216**	**4,872**	**5,659**
Total Liabilities & Shareholders Equity	**13,562**	**14,395**	**16,020**	**15,497**	**18,178**

Cash Flow					
OPERATING ACTIVITIES					
Net Income	**(997)**	**279**	**848**	**942**	**1,011**
Depreciation & Amort.	417	303	344	412	403
Amort. of Goodwill and Intangibles	27	21	28	42	47
Other Amortization	11	9	13	12	9
Add: Depreciation & Amort., Total	**455**	**333**	**385**	**466**	**459**
(Gain) Loss On Sale of Assets	5	(20)	(4)	73	(8)
(Gain) Loss On Sale Of Invest.	113	212	-	(5)	6
Total Asset Writedown	103	11	31	12	7
Provision for Credit Losses	11	-	-	-	-
(Income) Loss on Equity Invest.	(233)	(206)	(248)	(264)	(317)
Stock-Based Compensation	8	14	27	29	24
Other Operating Activities	1,673	344	356	322	198
Net Cash From Discontinued Ops.	(195)	282	233	(78)	(9)
Changes in working capital:					
Inventories	(50)	(28)	(51)	2	(5)
Account Receivables	(214)	(483)	276	169	(119)
Account Payables	(301)	498	(212)	(230)	140
Unearned Revenue	118	(17)	13	43	25
Other Net Operating Assets	(5)	-	(2)	-	-
Total changes working capital	**(452)**	**(30)**	**24**	**(16)**	**41**
Cash from Ops.	**491**	**1,219**	**1,652**	**1,481**	**1,412**
INVESTING ACTIVITIES					
Purchase of PPE and investment properties	(318)	(282)	(608)	(826)	(1,592)
Purchase/Sale of Intangibles	(15)	(8)	(7)	(10)	(16)
Capital Expenditures	**(333)**	**(290)**	**(615)**	**(836)**	**(1,608)**
Sale of PPE	61	47	2	5	17
Cash Acquisitions	(9)	-	(350)	(502)	(244)
Divestitures	54	-	-	-	(6)
Nuclear Fuel Expenditures	-	-	-	-	-
Cont. To Nuclear Decomm. Trust	-	-	-	-	-
Net Cash from Investments	(17)	18	(568)	(124)	(203)
Net (Increase)/Decrease in Loans Orig/Sold	-	-	-	-	-
Deferred Charges	-	-	-	-	-
Total Other Investing Activities	(1,037)	125	146	579	606
Cash from Investing	**(1,281)**	**(100)**	**(1,385)**	**(878)**	**(1,438)**
FINANCING ACTIVITIES					
Proceeds from borrowings	5,241	3,403	3,854	4,034	5,259
Total Debt Issued	**5,241**	**3,403**	**3,854**	**4,034**	**5,259**
Repayment of borrowings	(4,351)	(3,752)	(3,544)	(4,450)	(4,310)
Payment on lease liabilities	(28)	(15)	(23)	(21)	(25)
Total Debt Repaid	**(4,379)**	**(3,767)**	**(3,567)**	**(4,471)**	**(4,335)**
Repurchase of Common Stock	(16)	(13)	(27)	(61)	(19)
Common Dividends Paid	(872)	(107)	(125)	(160)	(250)
Special Dividend Paid	-	-	-	(71)	-
Other Financing Activities	81	(371)	(338)	(370)	(501)
Cash from Financing	**55**	**(855)**	**(203)**	**(1,099)**	**154**
CONSOLIDATED					
Cash and cash equivalents at the beginning of the year	1,740	1,009	1,297	1,246	732
Net (decrease) / increase in cash and cash equivalents	(735)	264	64	(496)	128
Foreign Exchange Rate Adj.	4	24	(79)	(18)	14
Misc. Cash Flow Adj.	-	-	(36)	-	(24)
Cash and cash equivalents at the end of the year	**1,009**	**1,297**	**1,246**	**732**	**850**
**Cash and cash equivalents in the consolidated statement of cash flows*					
Add: Restircted bank balances held as collateral by banks	23	47	8	35	21
Cash and cash equivalents at the beginning of the year	**1,032**	**1,344**	**1,254**	**767**	**871**
**Cash and cash equivalents in the balance sheets*					

Appendix B - WACC Calculation for Normal Growth

Sembcorp Industries (STI: U96)
For the Financial Year Ending 31 December
All figures are expressed in S$ 000

Valuation Date	3/19/2025
Share Price (SGD)	6.20

Company Capital Structure		Source:
Current share price	$6.20	S&P Capital IQ
Total no. of shares outstanding (000)	######	S&P Capital IQ
Market capitalization (000)	$11,034	
Add: Total debt (Market value 000)	$9,400	S&P Capital IQ
Less: Total cash (000)	($985)	S&P Capital IQ
Add: Non-controlling interest (000)	$298	S&P Capital IQ
Enterprise value (000)	######	
		154%
		1,790,005

Revenue breakdown by geography			Source:	Growth:
Country	**Revenue (00**	**% of Total Revenu**	S&P Capital IQ	
Singapore	4,754,000	74.08%	50.00%	4.00%
UK	561,000	8.74%		
China	383,000	5.97%		
India	366,000	5.70%		
Rest of Asia	287,000	4.47%	50	8%
Middle East	48,000	0.75%		
Other Countries	18,000	0.28%		
Total	**6,417,000**	**100.00%**		

Weighted Average Cost of Capital (WACC)		Source:
Capital Asset Pricing Model (CAPM)		
Risk-free rate	2.69%	Singapore 10Y Government Bond Yield as at 3/19/2025
Equity risk premium	5.12%	Blended by Region (Damodaran)
Unlevered beta	0.50	Hamada Equation
Re-levered beta	0.86	Bottom-up Beta
Cost of equity	**7.07%**	
Cost of Debt		
Pre-tax cost of debt	4.60%	Weighted Average Cost of Debt as at 31/12/2024 (Company filings)
Country risk premium	0.49%	Blended by Region (Damodaran)
Corporate tax rate	17.00%	Singapore Corporate Tax Rate
After-tax cost of debt	**4.23%**	
Capital Structure		
Market D/E	85.19%	
Proportion of equity	54.00%	
Proportion of debt	46.00%	
WACC	**5.76%**	

Unblended				Source:
Countries	**Default Spre**	**Equity Risk Premiu**	**Country Risk Premi**	**Corporate Tax R**
Singapore	0.00%	4.33%	0.00%	17.00%
India	2.18%	7.26%	2.93%	30.00%
UK	0.53%	5.13%	0.80%	25.00%
China	0.70%	5.27%	0.94%	25.00%
Rest of Asia	1.14%	5.87%	1.54%	25.74%
Middle East	3.42%	4.03%	8.36%	13.46%
Other Countries	1.06%	5.76%	1.43%	25.35%

Source: Damodaran as at 1/8/2025

Blended				
Countries	**Default Spre**	**Equity Risk Premiu**	**Country Risk Premi**	**Corporate Tax Rate**
Singapore	0.00%	3.50%	0.00%	17.00%
India	0.13%	0.70%	0.26%	2.62%
UK	0.04%	0.31%	0.05%	1.43%
China	0.04%	0.30%	0.05%	1.43%
Rest of Asia	0.05%	0.26%	0.07%	1.15%
Middle East	0.03%	0.03%	0.06%	0.10%
Other Countries	0.00%	0.02%	0.00%	0.07%
Total	**0.35%**	**5.12%**	**0.49%**	**23.86%**

Bottom Up Beta Computation				
Company	**Country**	**Debt/Equity**	**Tax Rate**	**5Y Levered Beta**
Keppel Infrastructure Trust	Singapore	169.85%	17.00%	0.49
Union Gas Holdings Limited	Singapore	82.81%	17.00%	0.83
AGL Energy Limited	Australia	44.18%	25.35%	0.37
Origin Energy Limited	Australia	92.29%	25.35%	1.23
ENN Energy Holdings Limited	Hong Kon	45.26%	25.74%	0.79
Adani Total Gas Limited	India	42.84%	30.00%	1.36
YTL Power International Berhad	Malaysia	196.38%	25.74%	1.05
Shenzhen Gas Corporation Ltd.	China	80.82%	25.00%	0.41
Median		**81.82%**	**25.35%**	**0.81**

2028 Forecasted GDP Growth Rate		% of Total Revenu
Singapore	2.80%	74.08%
UK	1.50%	8.74%
India	6%	5.97%
China	3.60%	5.70%
Vietnam	6.50%	4.47%
UAE	3.10%	0.75%
Oman	1.90%	0.28%

Appendix C - WACC Calculation for High Growth

Sembcorp Industries (STI: U96)
For the Financial Year Ending 31 December
All figures are expressed in S$ 000

Valuation Date	3/19/2025
Share Price (SGD)	6.20

Company Capital Structure		Source:
Current share price	$6.20	S&P Capital IQ
Total no. of shares outstanding (000)	######	S&P Capital IQ
Market capitalization (000)	$11,034	
Add: Total debt (Market value 000)	$9,400	S&P Capital IQ
Less: Total cash (000)	($985)	S&P Capital IQ
Add: Non-controlling interest (000)	$298	S&P Capital IQ
Enterprise value (000)	######	
		154%
		1,790,005

Revenue breakdown by geography			Source:	Growth:
Country	**Revenue (00**	**% of Total Revenu**	S&P Capital IQ	
Singapore	4,754,000	74.08%	50.00%	4.00%
UK	561,000	8.74%		
China	383,000	5.97%		
India	366,000	5.70%		
Rest of Asia	287,000	4.47%	50	8%
Middle East	48,000	0.75%		
Other Countries	18,000	0.28%		
Total	**6,417,000**	**100.00%**		

Weighted Average Cost of Capital (WACC)		Source:
Capital Asset Pricing Model (CAPM)		
Risk-free rate	2.69%	Singapore 10Y Government Bond Yield as at 3/19/2025
Equity risk premium	5.05%	Blended by Region (Damodaran)
Unlevered beta	0.50	Hamada Equation
Re-levered beta	0.86	Bottom-up Beta
Cost of equity	**7.02%**	
Cost of Debt		
Pre-tax cost of debt	7.00%	Project Cost of Debt
Country risk premium	0.49%	Blended by Region (Damodaran)
Corporate tax rate	17.00%	Singapore Corporate Tax Rate
After-tax cost of debt	**6.22%**	
Capital Structure		
Market D/E	85.19%	
Proportion of equity	54.00%	
Proportion of debt	46.00%	
WACC	**6.65%**	

Unblended				Source:
Countries	**Default Spre**	**Equity Risk Premiu**	**Country Risk Premi**	**Corporate Tax R**
Singapore	0.00%	4.33%	0.00%	17.00%
India	2.18%	7.26%	2.93%	30.00%
UK	0.53%	5.13%	0.80%	25.00%
China	0.70%	5.27%	0.94%	25.00%
Rest of Asia	1.14%	5.87%	1.54%	25.74%
Middle East	3.42%	4.03%	8.36%	13.46%
Other Countries	1.06%	5.76%	1.43%	25.35%

Source: Damodaran as at 1/8/2025

Blended				
Countries	**Default Spre**	**Equity Risk Premiu**	**Country Risk Premi**	**Corporate Tax Rate**
Singapore	0.00%	3.50%	0.00%	12.59%
India	0.13%	0.63%	0.26%	2.62%
UK	0.04%	0.31%	0.05%	1.43%
China	0.04%	0.30%	0.05%	1.43%
Rest of Asia	0.05%	0.26%	0.07%	1.15%
Middle East	0.03%	0.03%	0.06%	0.10%
Other Countries	0.00%	0.02%	0.00%	0.07%
Total	**0.35%**	**5.05%**	**0.49%**	**19.46%**

Bottom Up Beta Computation				
Company	**Country**	**Debt/Equity**	**Tax Rate**	**5Y Levered Beta**
Keppel Infrastructure Trust	Singapore	169.85%	17.00%	0.49
Union Gas Holdings Limited	Singapore	82.81%	17.00%	0.83
AGL Energy Limited	Australia	44.18%	25.35%	0.37
Origin Energy Limited	Australia	92.29%	25.35%	1.23
ENN Energy Holdings Limited	Hong Kon	45.26%	25.74%	0.79
Adani Total Gas Limited	India	42.84%	30.00%	1.36
YTL Power International Berhad	Malaysia	196.38%	25.74%	1.05
Shenzhen Gas Corporation Ltd.	China	80.82%	25.00%	0.41
Median		**81.82%**	#####	**0.81**

2028 Forecasted GDP Growth Rate		% of Total Revenu
Singapore	2.80%	74.08%
UK	1.50%	8.74%
India	6%	5.97%
China	3.60%	5.70%
Vietnam	6.50%	4.47%
UAE	3.10%	0.75%
Oman	1.90%	0.28%

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